WASP-12b

R-0544

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_181113` · created 2026-07-22T04:17:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458435.90559 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1112 +/- 0.0075
Transit depth (Rp/Rs)^2	1.24 +/- 0.17 [%]
Orbital Inclination (inc)	84.86 +/- 2.06
Airmass coefficient 1 (a1)	1.1593 +/- 0.0089
Airmass coefficient 2 (a2)	-0.134 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	None
Optimal Aperture	4.89
Optimal Annulus	11.45
Transit Duration (day)	0.1258 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1112 ± 0.0075 vs 0.1178 (deviation 0.79σ)
Duration fit vs published	0.1258 d vs 0.125 d (deviation 0.22σ)
Mid-time O–C	0.9 min
Depth SNR	13.3
Residual scatter	0.77 %

Light curve

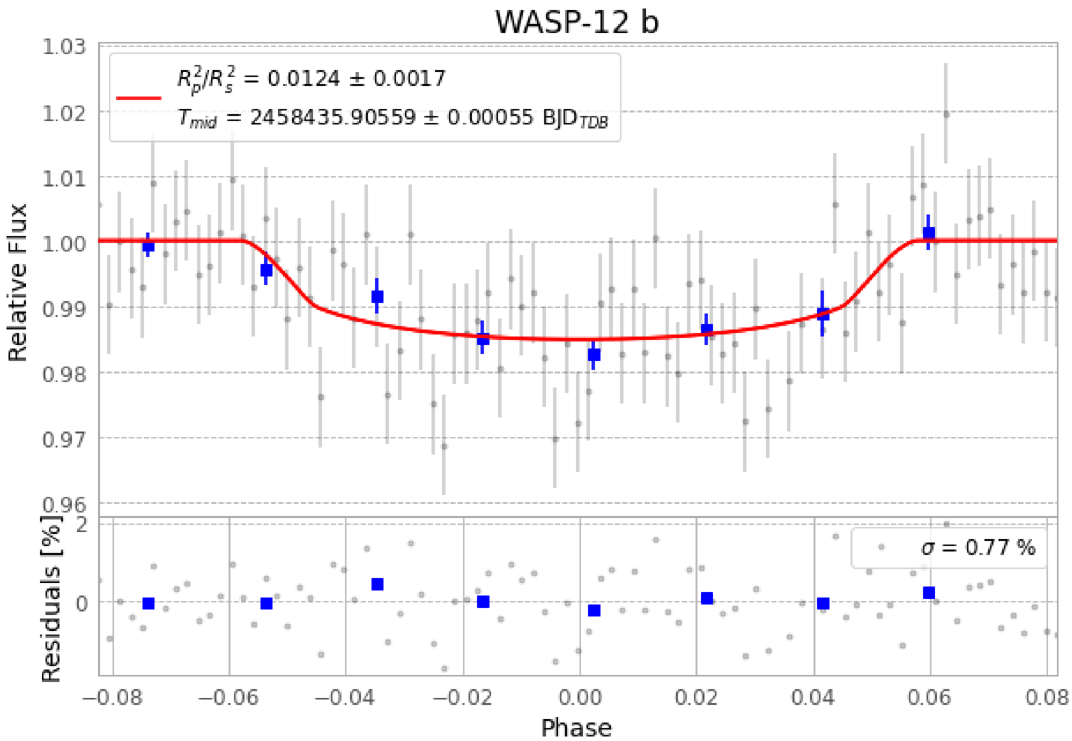

Figure 1 — FinalLightCurve_WASP-12 b_2018-11-13.png (SHA-256 6d9f48d90fd93ad3…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 218], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 33bf488bd17ad5b6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

87 FITS frames (57 MB), `WASP-121811113072712.FITS` → `WASP-121811113114512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-181113.log (470ef9b1ecf1…), FinalLightCurve_WASP-12 b_2018-11-13.png (6d9f48d90fd9…), FinalLightCurve_WASP-12 b_2018-11-13.csv (9ad4187441c4…)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)